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MA 324
STATISTICAL INFERENCE AND MULTIVARIATE ANALYSIS
IIT GUWAHATI

QUIZ III
16:05–16:50 IST
APRIL 08, 2026

Answers must be given *exclusively on this sheet*: answers given on the other sheets *will be ignored*. This question paper has **6 questions** and **2 pages**.

NAME AND ROLL NUMBER

Please mark your Roll No:

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230123019

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[Signature]

QUESTIONS AND RESPONSES

Question 1: (3 points) For $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, and $\beta \in \mathbb{R}^p$, consider a regression model

$$Y = X\beta + \epsilon,$$

where ϵ has an n -dimensional multivariate normal distribution with zero mean vector and identity covariance matrix. Let I_p denote the identity matrix of order p . For $\lambda > 0$, let

$$\hat{\beta}_n = (X^T X + \lambda I_p)^{-1} X^T Y,$$

be an estimator of β . Then which of the following options is/are correct?

☒ $\hat{\beta}_n$ has a multivariate normal distribution

☒ $\text{Var}(\hat{\beta}_n) = (X^T X + \lambda I_p)^{-1}$

☒ $\hat{\beta}_n$ is an unbiased estimator of β

☒ $(X^T X + \lambda I_p)$ is a positive definite matrix

Question 2: (3 points) Consider the multiple linear regression problem

$$Y = X\beta + \epsilon, \text{ and } \hat{Y} = X\hat{\beta}$$

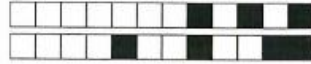
where $Y = (Y_1, \dots, Y_n)'$ is a $n \times 1$ vector of response variable, X is a $n \times p$ design matrix with full rank (largest possible rank), β is a $p \times 1$ vector of parameters, and ϵ is a $n \times 1$ vector of errors; $n > p$. The first element of β is the intercept of the model. Then which of the following statements is/are true?

☒ $\sum_{i=1}^n \text{Var}(\hat{Y}_i) = \sigma^2 p$

☒ $\sum_{i=1}^n \text{Var}(\hat{Y}_i) < \sigma^2 p$

☐ $\sum_{i=1}^n \text{Var}(\hat{Y}_i) = \sigma^2 n$

☒ $\sum_{i=1}^n \text{Var}(\hat{Y}_i) < \sigma^2 n$



Question 3: (3 points) Let $y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \cdots + \beta_{22} x_{22,i} + \epsilon_i, i = 1, 2, \dots, 123$, where, $x_{1,i}, \dots, x_{22,i}$ are fixed input (independent) variables; and, for $j = 0, 1, 2, \dots, 22$; β_j are unknown parameters, and ϵ_i 's are independent and identically distributed normal random variables with mean zero and finite variance. If $SS_{Reg} = 338.92$ and $SS_T = 522.30$ and the value of R_{adj}^2 (adjusted R^2) equals to t . Then which of the following statements is/are true?

☐ $0.43 < t < 0.54$

☒ $0.51 < t < 0.59$

☒ $0.61 < t < 0.69$

☒ $0.53 \leq t < 0.63$

Question 4: (2 points) Let a linear model $Y = \beta_0 + \beta_1 X + \epsilon$ be fitted to the following data, where ϵ is normally distributed with mean 0 and unknown variance $\sigma^2 > 0$.

x_i	0	1	2	3	4
y_i	3	4	5	6	7

Let $\hat{Y}_0$ denote the ordinary least-square estimator of Y at $X = 6$. If $E(\hat{Y}) = c_0\beta_0 + c_1\beta_1$ and $\text{Var}(\hat{Y}_0) = d\sigma^2$, then which of the following is/are true?

☐ $c_1 = 6, c_2 = 1, 1.7 \leq d \leq 1.9$

☒ $c_1 = 1, c_2 = 6, 1.7 \leq d \leq 1.9$

☒ $3.1 \leq c_2 - c_1 - d \leq 3.5$

☒ $c_1 + c_2 + d \leq 8.8$

Question 5: (2 points) Consider a multiple linear regression problem. Which of the following statements is/are true?

☒ A linear model is linear in the parameters, but not necessarily in the independent variables

☒ Q-Q plot is used to test for normality

☒ A leverage point is always influential point

☒ PRESS is a measure of how well a regression model perform in predicting new observations

Question 6: (2 points) Consider the linear model

$$Y_1 = \beta_1 + \beta_2 + \epsilon_1$$

$$Y_2 = \beta_1 + 2\beta_2 + \epsilon_2$$

$$Y_3 = \beta_1 + c\beta_2 + \epsilon_3,$$

where $\beta_1, \beta_2 \in \mathbb{R}$ are unknown parameters, and the uncorrelated errors $\epsilon_i, i = 1, 2, 3$ have zero mean and finite variance $\sigma^2 (> 0)$. The constant c is such that $\hat{\beta}_1$ and $\hat{\beta}_2$ are uncorrelated, where $\hat{\beta}_1$ and $\hat{\beta}_2$ are the best linear unbiased estimators of β_1 and β_2 , respectively. Then $(\text{Var}(\hat{\beta}_1), \text{Var}(\hat{\beta}_2))$ equals

☐ $(3\sigma^2, \frac{\sigma^2}{14})$

☐ $(\frac{\sigma^2}{14}, \frac{\sigma^2}{3})$

☐ $(\frac{\sigma^2}{3}, 14\sigma^2)$

☒ $(\frac{\sigma^2}{3}, \frac{\sigma^2}{14})$